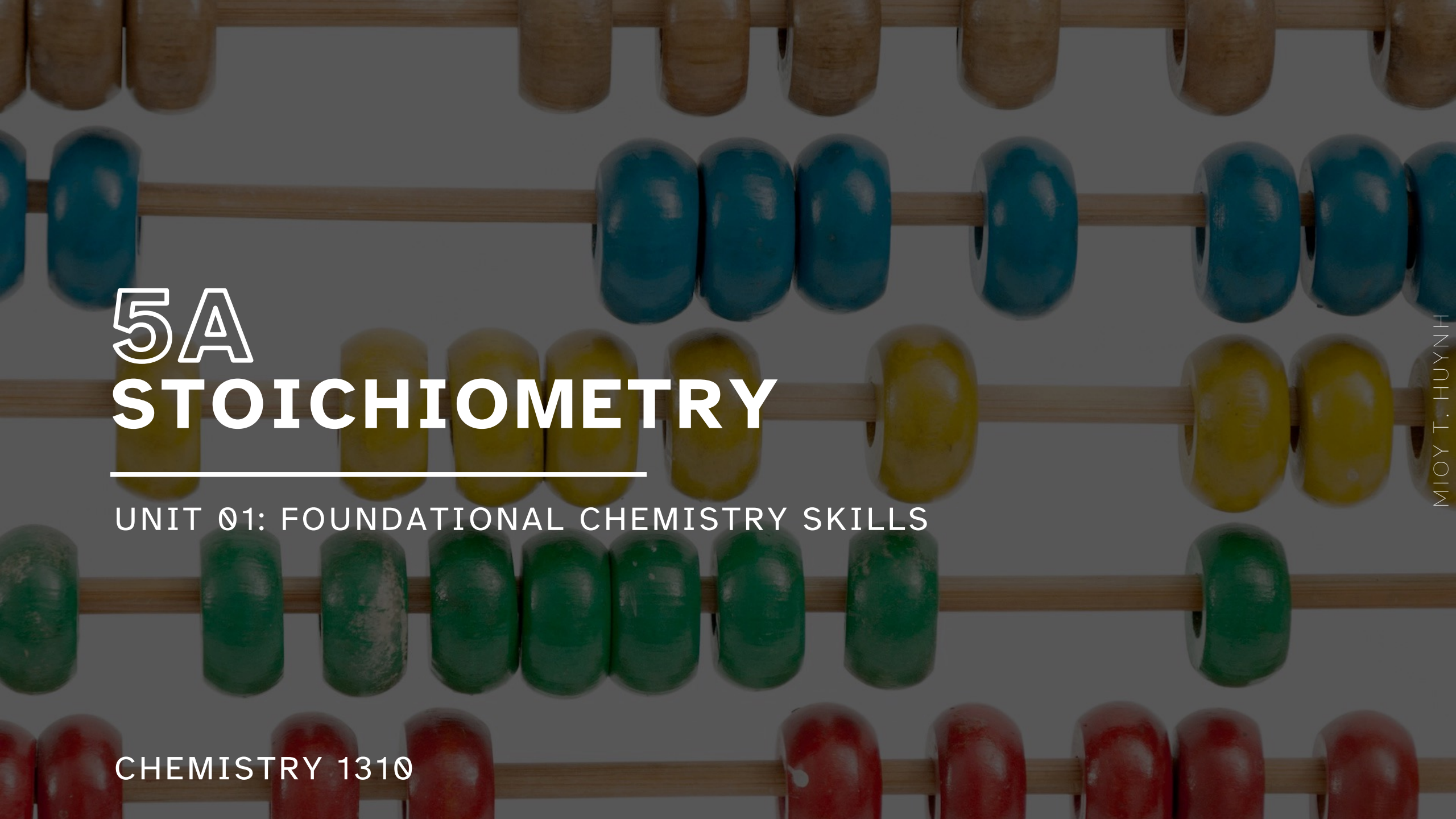




5A STOICHIOMETRY

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 5

1. Calculate the number of moles of any substance involved in a chemical reaction from the number of moles of any other substance in the reaction.
2. Use the mass of one substance to determine the masses of other substances involved in a chemical reaction.
3. Calculate the quantities of substances produced in a reaction when quantities of more than one reactant are specified.
4. Express the quantity of product obtained from a reaction as a percentage of what the reaction is theoretically capable of producing.
5. Solve problems using molarity.
6. Extend the concept of molarity to the individual ions in solutions of ionic substances.
7. Determine how much of any substance is involved in a chemical reaction, given any quantity of another substance, no matter what units are involved.

Mole Calculations for Chemical Reactions

5.1

Stoichiometry is a term used to describe *quantitative* relationships in chemistry.

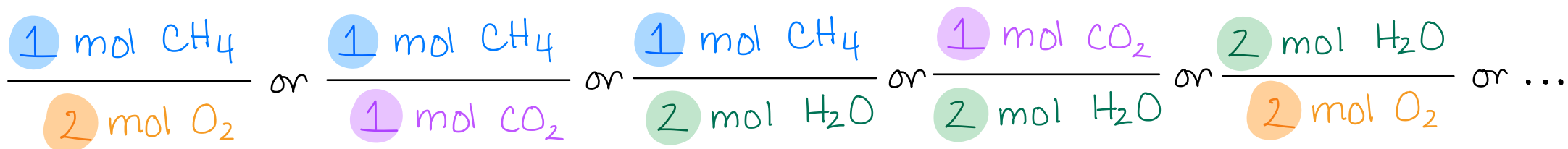
- Any chemistry question that asks “*How much...?*” of a particular substance is consumed or formed in a chemical reaction is a stoichiometry problem.

All stoichiometry problems must start with a balanced chemical equation.

- Stoichiometric coefficients from balanced chemical equations provide the mole-to-mole ratios that are used to construct conversion factors between one substance to another in the reaction.



any two species

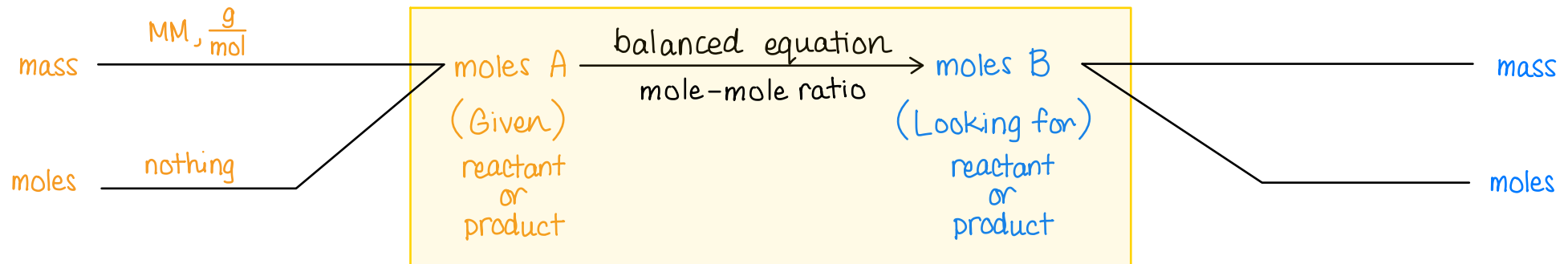


Mass Calculations for Chemical Reactions

5.2

It is impossible to count a mole of substance.

- Use molar masses (g/mol) to convert from mass to moles or vice versa.



IN-CLASS QUESTION 1

Learning Objective(s): 5.1

How many moles of O_2 are required to produce 12.5 mol of NO_2 according to the *unbalanced* reaction below?

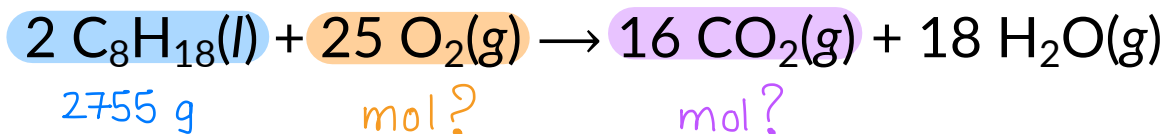


$$12.5 \text{ mol } NO_2 \times \frac{7 \text{ mol } O_2}{4 \text{ mol } NO_2} = 21.9 \text{ mol } O_2$$

mole-mole ratio

EXAMPLE PROBLEM

Learning Objective(s): 5.1, 5.2



$mm = 114.232 \text{ g/mol}$

How many moles of oxygen are required to react with 2755 g of octane, C_8H_{18} ?

mass $\xrightarrow{\text{molar mass}}$ moles $\xrightarrow{\text{mole ratio}}$ moles
 C_8H_{18} C_8H_{18} O_2

$$2755 \text{ g C}_8\text{H}_{18} \times \frac{1 \text{ mol C}_8\text{H}_{18}}{114.232 \text{ g C}_8\text{H}_{18}} = 24.117 \text{ mol C}_8\text{H}_{18} \times \frac{25 \text{ mol O}_2}{2 \text{ mol C}_8\text{H}_{18}} = 301.5 \text{ mol O}_2 \times \frac{16 \text{ mol CO}_2}{25 \text{ mol O}_2} = 192.9 \text{ mol CO}_2$$

How many moles of carbon dioxide are produced?

$$\underset{\text{mass}}{2755 \text{ g C}_8\text{H}_{18}} \times \underset{\text{molar mass}}{\frac{1 \text{ mol C}_8\text{H}_{18}}{114.232 \text{ g C}_8\text{H}_{18}}} = \underset{\text{moles}}{24.117 \text{ mol C}_8\text{H}_{18}} \times \underset{\text{mole ratio}}{\frac{16 \text{ mol CO}_2}{2 \text{ mol C}_8\text{H}_{18}}} = 192.9 \text{ mol CO}_2$$

multiple pathways

Problems Involving Limiting Quantities

5.3

In most reactions, one reactant will run out before the others.

- The reactant that runs out first is called the limiting reactant (**LR** or limiting reagent) because it *limits* the amount of products formed.
- The other reactants are called the excess reactants (**xs** or excess reagents).

For example, consider the reaction between nitrogen and hydrogen to form ammonia.

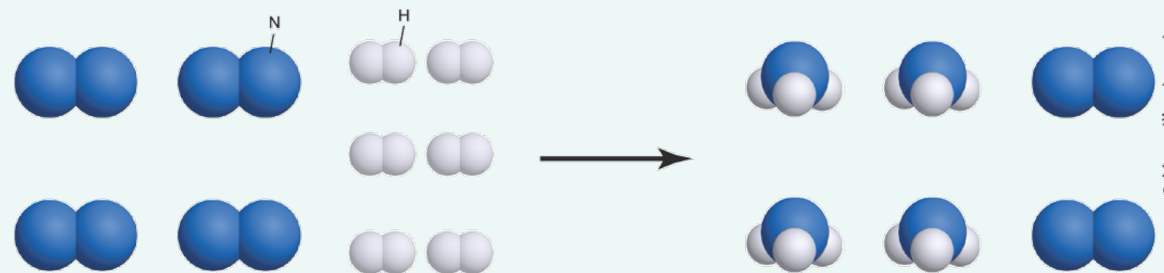
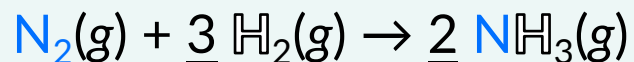
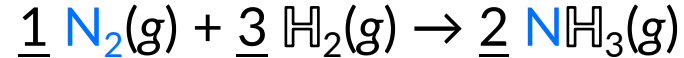


FIGURE 5.4

- If the reaction starts with **4 molecules of N₂** and 6 molecules of H₂, then **H₂** is the **LR**.
- At the *end* of the reaction: H₂ runs out first (*completely* consumed), only 4 molecules of **NH₃** are made, and **2 molecules of N₂** remain in excess.

Problems Involving Limiting Quantities

5.3



Conditions: The reaction started with 4 molecules of N_2 and 6 molecules of H_2 .

Let's ask two hypothetical questions to develop a method to identify the limiting (LR) and excess (xs) reactants in a chemical reaction.

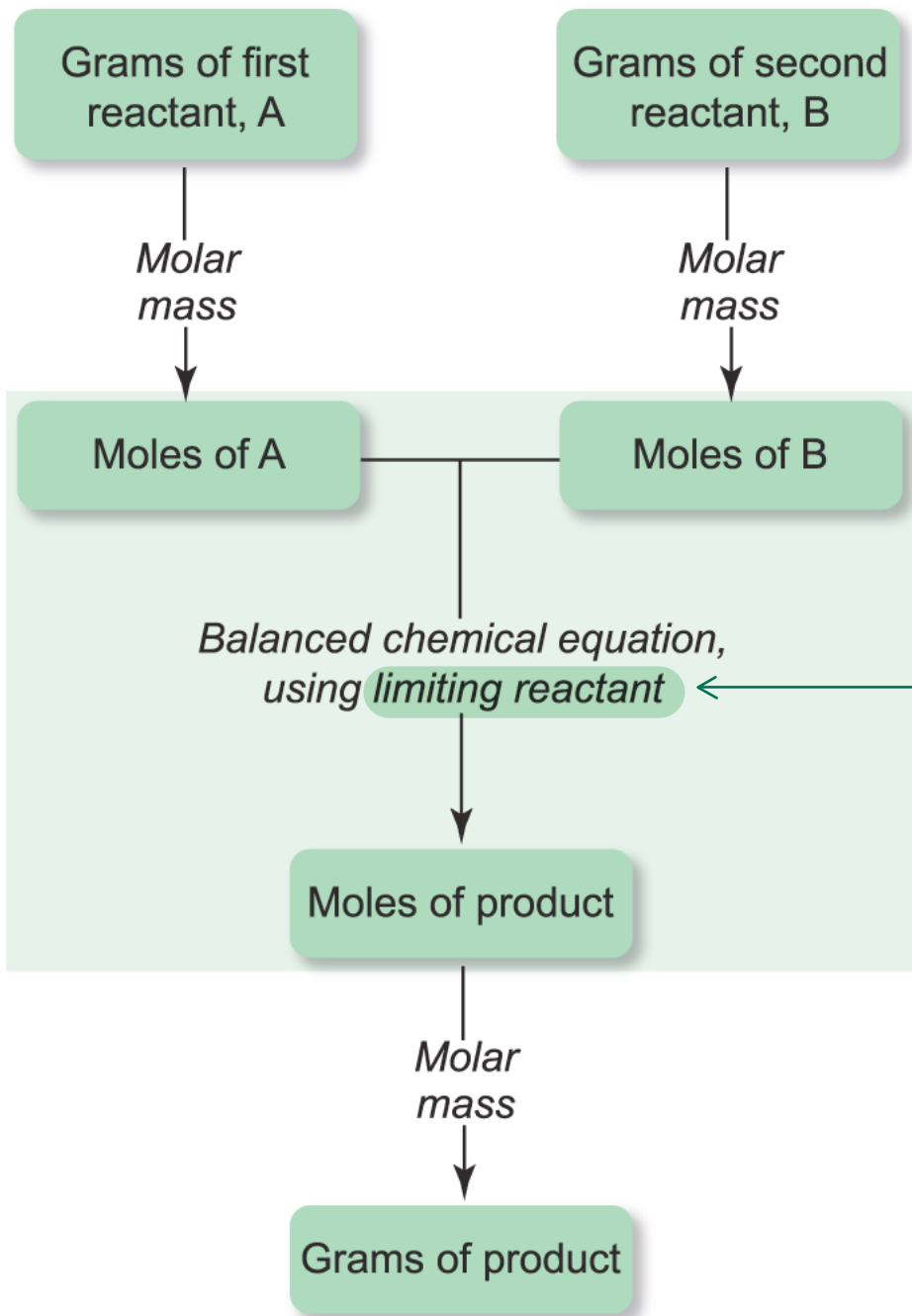
- How many molecules of NH_3 can be made using all 4 molecules of N_2 if infinite H_2 ?

$$4 \text{ molecules N}_2 \times \frac{2 \text{ molecules NH}_3}{1 \text{ molecules N}_2} = 8 \text{ molecules NH}_3$$

- How many molecules of NH_3 can be made using all 6 molecules of H_2 if infinite N_2 ?

$$6 \text{ molecules H}_2 \times \frac{2 \text{ molecules NH}_3}{3 \text{ molecules H}_2} = 4 \text{ molecules NH}_3$$

smaller amount of product,
so H_2 is limiting reactant (LR)



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Problems Involving Limiting Quantities

5.3

How to identify when a stoichiometry problem involves a limiting reactant?

- When the quantities or amounts of *all* reactants are given to you.

Q Which is LR?

A Calculate moles product from all reactant A and from all reactant B.

whichever reactant yields least product is LR!

FIGURE 5.7 Procedure for Limiting-Reactant Problems Involving Masses

IN-CLASS QUESTION 2

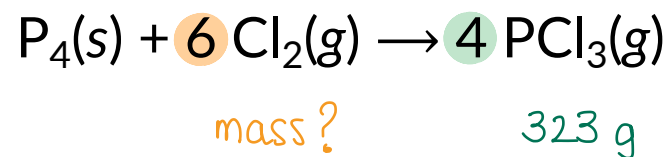
Learning Objective(s): 5.2

$$mm = 70.90 \text{ g/mol}$$

$$mm = 137.324 \text{ g/mol}$$

What mass of Cl_2 is required to produce 332 g PCl_3 in the presence of excess P_4 ?

balanced!



TIP This is not a LR problem!

mass $\text{PCl}_3 \xrightarrow{\text{mm}}$ moles $\text{PCl}_3 \xrightarrow{\text{ratio!}}$ moles $\text{Cl}_2 \xrightarrow{\text{mm}}$ mass Cl_2

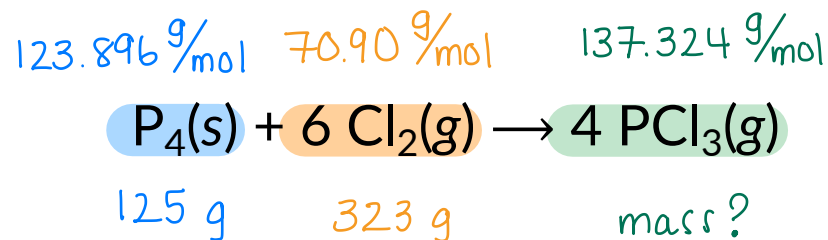
$$332 \text{ g PCl}_3 \times \frac{1 \text{ mol PCl}_3}{137.324 \text{ g PCl}_3} \times \frac{6 \text{ mol Cl}_2}{4 \text{ mol PCl}_3} \times \frac{70.90 \text{ g Cl}_2}{1 \text{ mol Cl}_2} = 257 \text{ g Cl}_2$$

IN-CLASS QUESTION 3

Learning Objective(s): 5.3

What is the *maximum* mass (in grams) of PCl_3 that can be produced from the reaction between 125 g P_4 and 323 g Cl_2 ?

TIP This is a LR problem because quantities of ALL reactants provided (i.e. they're limited)



$$125 \text{ g P}_4 \times \frac{1 \text{ mol P}_4}{123.896 \text{ g P}_4} \times \frac{4 \text{ mol PCl}_3}{1 \text{ mol P}_4} = 4.035 \text{ mol PCl}_3$$

$$323 \text{ g Cl}_2 \times \frac{1 \text{ mol Cl}_2}{70.90 \text{ g Cl}_2} \times \frac{4 \text{ mol PCl}_3}{6 \text{ mol Cl}_2} = 3.037 \text{ mol PCl}_3$$

Cl_2 is LR because it makes less product! (i.e. Cl_2 limits the rxn)

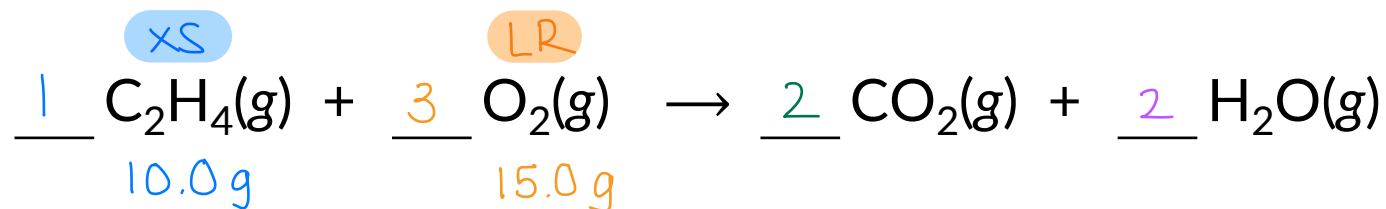
$$3.037 \text{ mol PCl}_3 \times \frac{137.324 \text{ g PCl}_3}{1 \text{ mol PCl}_3} = 417 \text{ g PCl}_3$$

EXAMPLE PROBLEM

Learning Objective(s): 5.3

TIP This is a LR problem because quantities of ALL reactants provided (i.e. they're limited)

If you have 10.0 g of C₂H₄ and 15.0 g O₂, which is the limiting reactant in the following reaction? How much of the excess reactant is consumed?



$$10.0 \text{ g C}_2\text{H}_4 \times \frac{1 \text{ mol C}_2\text{H}_4}{28.054 \text{ g C}_2\text{H}_4} \times \frac{2 \text{ mol CO}_2}{1 \text{ mol C}_2\text{H}_4} = 0.7129 \text{ mol CO}_2$$

$$15.0 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{31.998 \text{ g O}_2} \times \frac{2 \text{ mol CO}_2}{3 \text{ mol O}_2} = 0.3125 \text{ mol CO}_2$$

O₂ is LR because it makes less product!

$$15.0 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{31.998 \text{ g O}_2} \times \frac{1 \text{ mol C}_2\text{H}_4}{3 \text{ mol O}_2} \times \frac{28.054 \text{ g C}_2\text{H}_4}{1 \text{ mol C}_2\text{H}_4} = 4.38 \text{ g C}_2\text{H}_4 \text{ consumed!}$$

How much left?

$$\begin{array}{r} 10.0 \text{ g C}_2\text{H}_4 \\ - 4.38 \text{ g C}_2\text{H}_4 \\ \hline 5.6 \text{ g C}_2\text{H}_4 \end{array}$$

BONUS QUESTION

Assessed on: Fall 2021, Exam 1

What mass (in grams) of the excess reagent remains when 8.55 g of Si and 7.31 g of N₂ react according to the balanced chemical equation below?

